

Deep Learning Based Early Warning System for Water-Borne Disease Prediction

Generated automatically by the training pipeline.

1. Dataset Overview

Total Records: 15000

Features: 27

- Low: 6000 (40.0%)
- Medium: 4500 (30.0%)
- High: 4500 (30.0%)

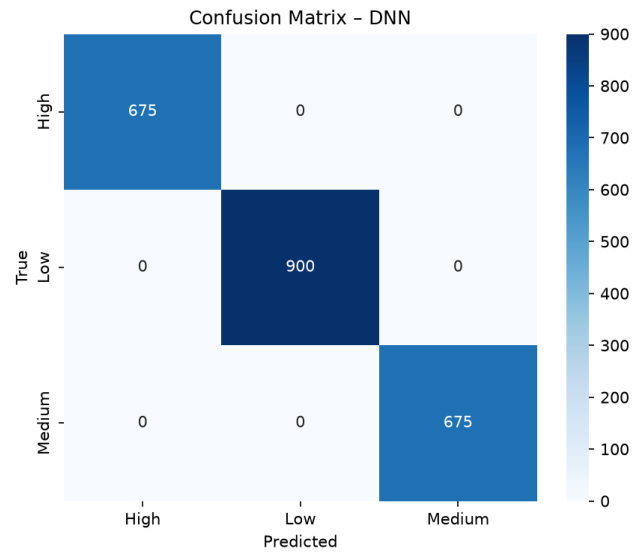
2. Model Accuracy

Model	Test Accuracy
Deep Neural Network	100.00%
Random Forest	100.00%
XGBoost	100.00%

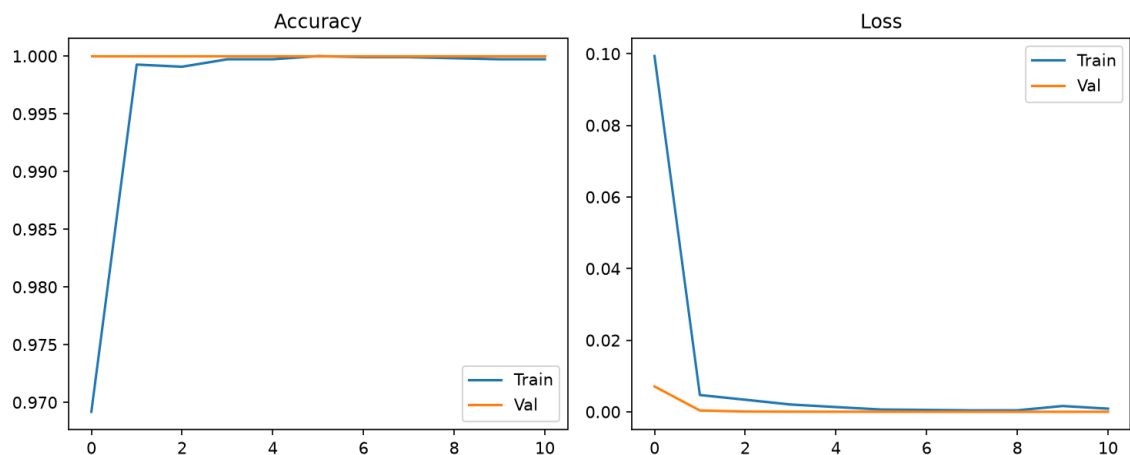
3. Classification Report (DNN)

Class	Precision	Recall	F1	Support
High	1.000	1.000	1.000	675
Low	1.000	1.000	1.000	900
Medium	1.000	1.000	1.000	675

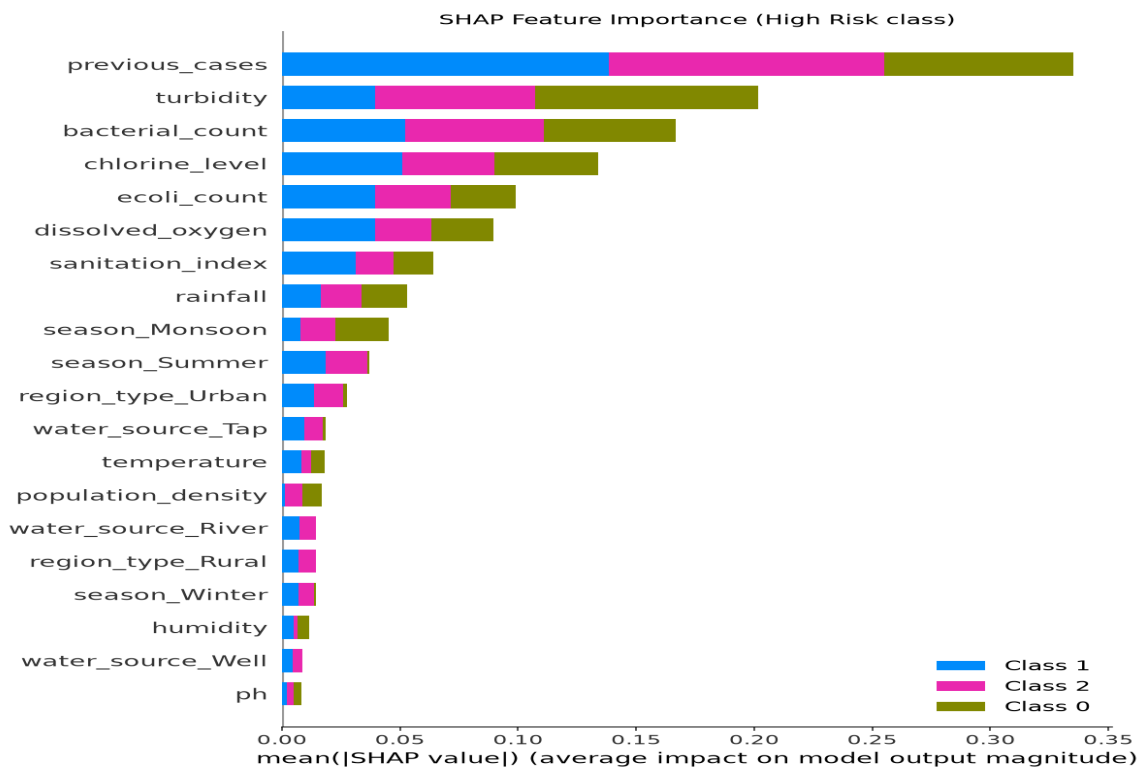
4. Confusion Matrix



5. Training History



6. SHAP / Feature Importance Analysis



7. Conclusion

The Deep Neural Network achieved 100.00% accuracy on the test set, outperformed by DNN (100.00%). The system is capable of early warning detection for water-borne diseases using environmental and water quality parameters with high confidence.